

16. Letter Frequency Attack on Monoalphabetic Substitution Cipher

```
... Enter Cipher Text: GSV XZHG
How many possible plaintexts do you want? 10

Top 10 Possible Plaintexts:
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Rank: 1
Score: 10
Plaintext: TAO INST

Rank: 2
Score: 10
Plaintext: OIN SHRO

Rank: 3
Score: 10
Plaintext: INS HRDI

Rank: 4
Score: 10
Plaintext: ETA OINE

Rank: 5
Score: 0
Plaintext: SHR DLCS
```

17. DES Decryption – Reverse Key Schedule

```
... DES Encryption Subkeys
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K1: 11100001100110010101010111111010101011001100111100011110
K2: 110000110011001010101011111110101010110011001111000111101
K3: 000011001100101010101111111110101011001100111100011110101
K4: 00110011001010101011111111000101100110011110001111010101
K5: 11001100101010101111111100000110011001111000111101010101
K6: 00110010101010111111110000111001100111100011110101010101
K7: 11001010101011111111000011000110011110001111010101010110
K8: 00101010101111111100001100111001111000111101010101011001
K9: 01010101011111111000011001100011110001111010101010110011
K10: 010101011111111100001100110011111000111101010101011001100
K11: 010101111111110000110011001011100011110101010101100110011
K12: 010111111111000011001100101010001111010101010110011001111
K13: 011111111100001100110010101010111101010101011001100111100
K14: 111111100001100110010101010111101010101100110011110001
K15: 111110000110011001010101011110101010110011001111000111
K16: 111100001100110010101010111101010101100110011110001111

DES Decryption Order
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K16 K15 K14 K13 K12 K11 K10 K9 K8 K7 K6 K5 K4 K3 K2 K1
```

18. DES – 28-bit Key Halves

```
print( 'D0 length: ', len(D0))

... Original Key:
133457799BBCDFF1

56-bit Key:
111100001100110010101010111101010101100110011110001111

C0 (28 bits):
1111000011001100101010101111

D0 (28 bits):
0101010101100110011110001111

C0 length: 28
D0 length: 28
```

19. CBC Mode Using 3DES

```
... Collecting pycryptodome
  Downloading pycryptodome-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
  Downloading pycryptodome-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
  2.3/2.3 MB 25.4 MB/s eta 0:00:00
Installing collected packages: pycryptodome
Successfully installed pycryptodome-3.23.0
Enter Plain Text: HELLO WORLD

IV: 24ba79896defc974
Key: 3d5e83c40dd35d6b0b70d3b0625bcd7fd5e66d8f1908977a
Ciphertext: 3a2a7ea8722699110bcf388f810d55cb
Decrypted Text: HELLO WORLD
```

20. ECB and CBC Error Propagation

```
... Original Plaintext:
b'Block-1-Data!!!Block-2-Data!!!Block-3-Data!!!\x03\x03\x03'

Original Ciphertext:
2527df24f0253ed54fd15d83e9c6b2d438e7d862d92f14c7c8e258b6fd6503fdc20a641b8779c23bda940d9f7804c7df

Modified Ciphertext:
2427df24f0253ed54fd15d83e9c6b2d438e7d862d92f14c7c8e258b6fd6503fdc20a641b8779c23bda940d9f7804c7df

Decrypted Result after Error:
b'!c\xa1\x14\xf39\x04\xc0\xda\x18\x14\x1c\x8b\x92nTmock-2-Data!!!Block-3-Data!!!\x03\x03\x03'
```